TOOKANG

Software Requirements Specification

Version 1.5 | March 2026 | Adds Solo Tukang vs Mandor/Crew Model

Platform:	Web (Next.js 14) + Mobile (React Native / Expo)
Language:	Bilingual — Bahasa Indonesia & English
Business Model:	Marketplace with Hybrid Escrow & Percentage Commission
AI Layer:	Claude API (Anthropic /v1/messages)
Target Market:	Bali — Homeowners, Expats, Villa & Property Managers
v1.5 Focus:	Introduces Solo Tukang (Freelancer) vs Mandor/Crew (Agency) model

Sections marked ★ NEW are new in v1.5 | ● SOLO = Solo-specific | ▲ CREW = Mandor/Crew-specific

1. Project Overview

Tookang is a bilingual (Bahasa Indonesia / English) marketplace platform connecting homeowners, expats, and property managers in Bali with trusted local craftsmen (tukang). The platform operates as a managed marketplace with a hybrid escrow model and now supports two distinct service-provider types — Solo Tukang (individual craftsmen) and Mandor/Crew (foreman-led teams) — mirroring the Upwork Freelancer vs Agency model.

1.1 The Core Problem

- No central directory of verified tradespeople in Bali
- Language barrier between expats and local Tukang
- No accountability — ghosting, incomplete jobs, price disputes are common
- No way to compare individual craftsmen vs trade crews before hiring
- Complex multi-trade jobs (e.g. full villa renovation) have no structured quoting path

1.2 The Solution

- Searchable directory of Solo Tukang and Mandor/Crew teams across 8 trade categories
- Two-stage bidding (Inspection → Final Quote) with job-size routing to appropriate tier
- AI-powered job assistant, photo analysis, voice input, and bilingual translation
- Hybrid escrow with milestone-based fund release
- Fully independent rating systems — solo rating and agency rating never cross-pollinate

2. Goals & Success Metrics

Target: 3 months post-launch

Goal	Metric	Target (3 months)
Grow Solo Tukang supply	Registered & verified Solo Tukang	200+ in Bali
Grow Mandor/Crew supply	Registered & verified Mandor crews	30+ in Bali
Drive demand	Jobs posted per week	100+
Ensure quality	Average rating (both tiers)	≥ 4.2 / 5.0
Monetize	Completed escrow transactions	50+ / month
Large-job capture	Jobs taken by Mandor/Crew	> 20% of total
Trust	Dispute rate	< 5% of completed jobs
Retention	User repeat hire rate	> 30%

3. User Roles

3.1 User (Job Poster)

A homeowner, expat, villa manager, or anyone who needs a craftsman. They post jobs, receive bids from both Solo Tukang and Mandor/Crew teams, compare options, and hold payment in escrow.

3.2 Solo Tukang ● SOLO

An individual craftsman who works alone. Equivalent to an Upwork Freelancer. Has a personal profile, builds a solo reputation, and bids directly on jobs. Can optionally join a Mandor's crew later without losing their independent solo profile or rating history.

3.3 Mandor (Crew Lead) ▲ CREW ★ NEW

A foreman or contractor who manages a team of workers. Equivalent to an Upwork Agency Owner. The Mandor owns the agency profile, bids on jobs on behalf of the crew, and is solely accountable for job delivery. Individual crew members do not interact directly with Users during a crew-type job.

3.4 Crew Member ▲ CREW ★ NEW

A Tukang who belongs to one or more Mandor crews. They maintain a fully independent Solo Tukang profile and can bid on small jobs as an individual. When a Mandor assigns them to a crew job, they operate under the Mandor's accountability. Their solo reviews and agency-job internal scores are stored separately and never merged.

3.5 Admin

Tookang platform staff. They verify both Solo Tukang KTPs and Mandor business registrations, handle disputes, and monitor platform health across both tiers.

4. Functional Requirements

4.1 Authentication & Onboarding — Updated

Both Roles

- Register via email + password or Google OAuth
- Phone number verification (OTP via SMS/WhatsApp)
- Language preference (ID / EN) on first launch
- Role selection: "I need a Tukang" / "I am a Solo Tukang" / "I manage a crew (Mandor)"

User Onboarding

- Name, profile photo, location (area in Bali), optional property type

Solo Tukang Onboarding ● SOLO

- Personal info, WhatsApp number, service area, trade categories (max 3)
- KTP upload for identity verification
- Optional portfolio photos and daily/job rates
- Availability calendar setup
- Skill assessment quiz for Tookang Certified badge eligibility

Mandor Onboarding ▲ CREW ★ NEW

- Personal info + crew/agency brand name (e.g. "Bali Build Crew")
- WhatsApp number (used as the agency contact for bridge messages)
- Service area, trade categories covered by crew (can select all 8)
- KTP upload (Mandor personal) + NIB or SIUP business registration document
- Agency logo/photo and portfolio of past crew projects
- Invite crew members by phone number — they receive a WhatsApp invitation link
- Set crew size and list member trade specialties

4.2 User — Discovery & Job Posting — Updated

- Browse Solo Tukang and Mandor/Crew teams; toggle between views or show both
- Filter panel includes: Solo only / Crew welcome / Both
- AI Job Intelligence Assistant analyses description and suggests appropriate tier: if budget > Rp 5,000,000 or multiple trade categories detected → recommends opening to crews
- "Crew Welcome" toggle on job post (default ON for large/multi-trade jobs, OFF for small jobs)
- Voice input for job description (Bahasa Indonesia + English)
- Mandatory photo with AI analysis (relevance + severity)
- AI price estimate displayed before submission

4.3 Bidding — Updated

- A job may receive bids from Solo Tukang and/or Mandor/Crew depending on the Crew Welcome setting
- User's bid inbox clearly separates Solo bids and Crew bids with distinct card designs
- Solo bid card: individual photo, name, solo rating, badges, trade, rate
- Crew bid card: agency logo, crew name, agency rating, team member thumbnails assigned, multi-trade coverage, crew portfolio link
- User can compare both and accept one — accepting a crew bid locks out solo bids and vice versa

4.4 Guided Review Flow — Updated

- After job completion, User reviews the entity they hired — either the Solo Tukang or the Mandor/Crew as a unit
- Reviews for crew jobs go to the agency rating, never to individual crew members' solo profiles
- Mandor receives an internal crew member performance score (visible only to Mandor in their dashboard) — separate from public agency rating
- Guided questions: on time? price accuracy? would rehire? + optional free text

4.5 Job Completion Certificate (Auto-PDF)

- Auto-generated on confirmation for both solo and crew jobs
- For crew jobs, certificate lists: agency name + Mandor name + assigned crew members

4.6 Tukang Availability Calendar

- Solo Tukang: personal availability (Available Today / Busy / On Leave)
- Mandor: crew availability — can flag as available even if some members are busy, based on bench capacity

5. Solo Tukang vs Mandor/Crew — Full Specification

This section is the definitive reference for all logic that differs between the two service-provider tiers.
Where behaviour is identical, it is not repeated here — refer to the relevant section.
Design principle: ratings are ALWAYS kept separate. A Tukang's solo score and their agency-job internal score are stored in different columns and never averaged, combined, or displayed together.

5.1 Account Type Overview ★ NEW

Dimension	Solo Tukang ●	Mandor / Crew ▲
Upwork analogue	Freelancer	Agency
Profile owner	Individual Tukang	Mandor (foreman)
Trade categories	Max 3	All 8 (crew combined)
Job size	Small to medium	Medium to large / multi-trade
Bid as	Self	Agency brand
Accountable party	Solo Tukang	Mandor
Public rating	Solo rating (public)	Agency rating (public)
Internal score	None	Per crew member (Mandor-only)
Verification docs	KTP only	KTP + NIB/SIUP
Payout destination	Personal wallet	Agency wallet → Mandor distributes
Platform fee	10–15% (per badge tier)	10–15% (per agency tier)
Max active bids	10	20 (larger pipeline expected)
WhatsApp bridge	Tukang's personal number	Mandor's number (agency contact)
Can join crew	Yes, without losing solo profile	—

5.2 Profile Structure ★ NEW

Solo Tukang Profile ●

- Profile photo (personal)
- Full name
- Location & service area
- Trade categories (max 3)

Mandor / Agency Profile ▲

- Agency logo + crew brand name
- Mandor's name as contact
- Service area (all Bali or regional)
- Trade categories covered (crew-wide)

- Solo rating ★ (public)
- Solo badges (Top Rated, Certified...)
- Jobs completed (solo count only)
- Response rate & response time
- Portfolio photos (personal work)
- Hourly / daily / job rate
- Availability status
- KTP verified tick

- Agency rating ★ (public)
- Agency badges (separate set)
- Agency jobs completed
- Agency response rate
- Crew portfolio (larger projects)
- Project-based rate (not daily)
- Crew availability status
- NIB/SIUP verified tick
- Crew member thumbnails (clickable)

5.3 Crew Composition & Member Management ★ NEW

- A Mandor can have 2–50 active crew members
- Each crew member is a registered Solo Tukang on the platform — they have their own solo profile running in parallel
- Mandor invites crew members via phone number; the invitee receives a WhatsApp link to accept. Acceptance does not affect their solo profile
- Mandor assigns specific crew members to each job at bid time — e.g. "1 plumber + 1 electrician from our crew"
- Crew members can be part of multiple crews simultaneously (one Tukang, multiple Mandors)
- Mandor can remove a crew member at any time. Active jobs are not affected — the assignment stands
- Crew member can leave a crew at any time. Their solo profile is completely unaffected

5.4 Bid Card Design Differences ★ NEW

Solo Bid Card ●

- Tukang profile photo (large)
- Name + trade badge
- Solo star rating (e.g. ★ 4.8)
- Jobs completed: 47
- Inspection bid or direct quote amount
- Estimated duration
- Short bid message
- "View solo profile" link

Crew Bid Card ▲

- Agency logo (large)
- Crew name + verified agency badge
- Agency star rating (e.g. ★ 4.9)
- Agency jobs completed: 23
- Assigned members: 3 thumbnails
- Trade coverage icons for this job
- Project quote amount
- Estimated project duration
- "View agency profile" link

5.5 Rating System — Fully Separate ★ NEW

RULE: A Tukang's solo rating and their agency-job internal score are ALWAYS stored separately. They are never averaged, combined, displayed together, or used to infer the other.
This mirrors Upwork's model exactly — a freelancer's personal profile score does not change when they complete a job as part of an agency.

Rating Type	Who receives it	Where displayed	Who can see it
Solo public rating	Solo Tukang (when hired individually)	Solo profile · search results	Everyone
Agency public rating	Mandor/Crew (when agency is hired)	Agency profile · search results	Everyone
Crew member internal score	Individual crew member (per agency job)	Mandor's team dashboard only	Mandor only

- The crew member internal score is generated from the Mandor's private post-job rating of each team member they assigned — a 1–5 score + optional note
- This score is used by the Mandor to build a quality-sorted roster and decide who to assign to future jobs
- It is invisible to the User, to other Tukang, and to Admin (except in dispute investigation)
- If a crew member later bids solo on a job, only their solo rating is shown — crew internal scores are never surfaced

5.6 Job-Size Eligibility Routing ★ NEW

The AI Job Intelligence Assistant determines which tier(s) are eligible to bid based on job characteristics. This can be overridden by the User.

Job signal	AI recommendation	User can override?
Budget < Rp 1,000,000	Solo Tukang only	Yes — can open to crews
Budget Rp 1M – Rp 5M	Both tiers eligible	Yes
Budget > Rp 5,000,000	Crew recommended	Yes — can solo-only
Multi-trade categories detected	Crew recommended	Yes
requires_inspection = false + simple job	Solo preferred	Yes
requires_inspection = true + complex job	Both, crew highlighted	Yes

5.7 Verification Differences ★ NEW

Solo Verification ●	Agency Verification ▲

- KTP (national ID) upload
- Admin manual review (24–48 hrs)
- Optional: skill assessment quiz
- Skill pass → Certified badge eligible
- KTP verified tick on profile

- Mandor KTP upload
- NIB or SIUP business registration
- Both docs reviewed by Admin
- Agency portfolio spot-check
- Crew member count minimum: 2
- NIB/SIUP verified tick on agency profile
- No skill quiz (agency rated on project outcomes)

5.8 Platform Fee — Applied Equally by Tier ★ NEW

Tier / Badge	Platform Fee	Applies to
Tookang Certified Solo	10%	Solo jobs only
Top Rated Solo	12%	Solo jobs only
Standard Solo	15%	Solo jobs only
Verified Agency (NIB/SIUP)	12%	Crew jobs only
Unverified Agency	15%	Crew jobs only
Top Agency (new badge)	10%	Crew jobs only

Note: A Solo Tukang who also belongs to a crew is billed the crew rate for agency jobs and their solo rate for individual jobs — independently.

5.9 Payout Flow Differences ★ NEW

Solo Tukang Payout ●

- Escrow releases to personal wallet
- Solo Tukang withdraws to personal bank
- Min withdrawal: Rp 100,000
- Tookang is not involved in any labour split

Agency / Crew Payout ▲

- Escrow releases to agency wallet
- Mandor withdraws from agency wallet
- Mandor distributes to crew members externally (cash/transfer — not on-platform)
- Tookang is NOT responsible for internal crew payroll
- Min withdrawal: Rp 500,000 (higher minimum for agency)

6. Non-Functional Requirements

Requirement	Specification
Performance	Page load < 2s on 4G; AI responses streamed with loading state
Availability	99.5% uptime; Supabase managed infra
Security	Midtrans PCI-DSS; HTTPS everywhere; JWT auth; RLS on all tables
Data Privacy	UU PDP (Indonesian Data Protection Law) compliance
Localization	Full bilingual (ID/EN); IDR; WIB/WITA/WIT timezones
AI Fallback	If Claude API down, job posting falls back to standard manual form
Offline Tolerance	Key screens cached; WhatsApp Bridge functional on 2G
Accessibility	WCAG 2.1 AA minimum for web; ≥ 44pt touch targets on mobile
Rating isolation	Solo ratings and agency ratings stored in separate DB columns — no shared query path

7. AI Features Specification (Claude API)

All AI features use POST /v1/messages | Model: claude-sonnet-4-20250514
All prompts are bilingual (ID + EN). Every AI-dependent flow has a manual fallback.

7.1 Job Intelligence Assistant — Updated

Now also determines which tier to recommend (solo / crew / both) as part of its structured output.

- Output JSON: { category, title, budget_range, urgency, requires_inspection, missing_fields[], recommended_tier: 'solo'|'crew'|'both' }
- If multi-trade categories are detected in one description, recommended_tier = 'crew'

7.2 Smart Tukang Matching — Updated

Runs twice per job — once to rank Solo Tukang, once to rank Mandor/Crew. Results are served as two separate sorted lists to the User.

Factor	Weight	Solo source	Crew source
Skill / Category match	30%	tukang_profiles.categories	agencies.trade_categories
Distance from job	20%	tukang_profiles.service_area	agencies.service_area
Rating average	20%	solo_rating_avg	agency_rating_avg
Jobs completed	15%	solo_jobs_count	agency_jobs_count
Response rate	10%	messages avg (solo thread)	messages avg (agency thread)
Availability	5%	availability_status	crew_availability_status

7.3 Bilingual Chat Bridge

Identical for both tiers. For crew jobs, the translation bridge connects the User's in-app chat to the Mandor's WhatsApp — not to individual crew members.

7.4 Trust & Safety

Flags suspicious reviews across both rating systems independently.

7.5 Dispute Assistant

Unchanged — Mandor is the accountable party for crew jobs, so dispute resolution is always with the Mandor.

7.6 Voice Input

Unchanged.

7.7 AI Price Estimator

Generates two estimates for eligible jobs: one for solo hire and one for crew hire, shown side by side.

7.8 AI Photo Analysis

Unchanged.

8. Payment, Escrow & Bidding Specification

The core two-stage escrow logic is identical for both tiers. Differences are noted below.

8.1 Platform Fee Logic

See Section 5.8 for the full fee table by tier and badge.

8.2 Two-Stage Bidding & Escrow Flow

Stages are identical for solo and crew jobs. The only difference is who submits what:

- Stage 1 (Inspection Bid): submitted by Solo Tukang / Mandor
- GPS check-in: Solo Tukang checks in personally. For crew jobs, the Mandor checks in (or designates a crew lead to check in using the Mandor's app)
- Stage 2 (Final Quote): submitted by Solo Tukang / Mandor
- Material Deposit: up to 30% released on Job Start photo — same for both tiers
- Job Finish photo: submitted by Solo Tukang / Mandor
- Auto-release: 72 hours after Job Finish photo — same for both

8.3 Bid Rules & Cancellation Policy

- Solo Tukang: max 10 active bids at any time
- Mandor/Crew: max 20 active bids (larger pipeline expected for agencies)
- Bid expiry, retraction, and cancellation rules are identical across both tiers

8.4 Dispute Resolution

Identical process for both tiers. For crew jobs, the Mandor is the sole accountable party — crew members are not named in disputes or resolution outcomes.

9. Tukang Wallet & Payout Specification

9.1 Solo Tukang Wallet ●

- Accumulates earnings from solo jobs only
- Available balance, pending balance, material advance tracking
- Min withdrawal: Rp 100,000 to personal bank account
- Payout via Midtrans Payouts (MVP) / Xendit (post-launch)

9.2 Agency Wallet ▲ ★ NEW

- Separate wallet entity linked to the agency, not the Mandor's personal profile
- Accumulates earnings from crew/agency jobs only
- Min withdrawal: Rp 500,000 to Mandor's registered business bank account
- Payout displayed as: agency jobs completed × net amount after platform fee
- Mandor's personal solo wallet (if they also do solo work) is a completely separate balance
- Tookang does not facilitate or track internal crew payroll — that is outside platform scope

10. Notification System

10.1 Notification Channels

Event	User	Solo Tukang	Mandor (Crew)
New bid received	Push + in-app	—	—
Bid accepted	—	Push + WhatsApp	Push + WhatsApp
Escrow funded	In-app	Push + WhatsApp	Push + WhatsApp
Final quote submitted	Push + in-app	—	—
Job Start photo uploaded	Push + in-app	—	—
Job Finish photo	Push + in-app	—	—
Funds released to wallet	—	Push + WhatsApp	Push + WhatsApp (agency wallet)
Dispute opened	Push + email	Push + WhatsApp	Push + WhatsApp
Crew invite received	—	WhatsApp	—

10.2 Implementation

- Push: Expo Push (mobile) + browser push (web)
- WhatsApp: Twilio/Wati API — Mandor's number for all crew-job notifications
- SMS fallback for critical events (escrow funded, dispute)
- All notifications respect language preference (ID / EN)

11. Tukang Label & Badge System — Updated

Two independent badge sets. Solo badges only appear on solo profiles. Agency badges only appear on agency profiles. A Tukang who belongs to a crew shows their solo badges when browsing in solo mode.

11.1 Solo Tukang Badges ●

Badge	Emoji	Criteria
Newly Joined	🟢	< 30 days, ≥ 70% profile completion — solo only
Top Rated	🟡	≥ 10 solo jobs, ≥ 4.5 solo rating, ≥ 80% response rate
Tookang Certified	🟣	KTP verified, skill assessment passed, ≥ 20 solo jobs, zero solo disputes. Highest search boost + 10% fee.
Repeat Favourite	🟠	Rehired by ≥ 3 unique Users (solo contracts only)
Fast Responder	🔵	Average solo first-reply < 1 hour
Rising Star	🟩	< 90 days, ≥ 5 solo jobs, recent solo ratings ≥ 4.5

11.2 Agency / Mandor Badges ▲ ★ NEW

Badge	Emoji	Criteria
New Agency	🟢	< 60 days on platform, NIB/SIUP verified, ≥ 2 crew members
Top Agency	🟡	≥ 10 agency jobs, ≥ 4.5 agency rating — earns 10% fee tier
Certified Agency	🟣	NIB/SIUP + KTP verified, ≥ 20 agency jobs, zero agency disputes, Admin-approved portfolio review
Multi-Trade Crew	🟠	Has verified members across ≥ 4 distinct trade categories
Fast Mandor	🔵	Average agency bid response < 2 hours (crew response budgeted longer than solo)
Rising Crew	🟩	< 120 days, ≥ 5 agency jobs, recent ratings ≥ 4.5

12. In-App Chat & WhatsApp Bridge

12.1 Core In-App Chat

- Real-time text, read receipts, media uploads (Supabase Realtime)
- Server-side bilingual translation on every message
- Mandatory Job Start + Job Finish photo prompts injected as system messages
- For crew jobs: chat thread is User ↔ Mandor only. Crew members do not have a chat channel with the User

12.2 WhatsApp Bridge

- Solo jobs: User in-app messages → Solo Tukang's WhatsApp
- Crew jobs: User in-app messages → Mandor's WhatsApp (the agency contact number)
- Media sync: photos from either WhatsApp number are auto-attached to the correct job record

12.3 Deep-Link Specification

Event	Deep link target
New job match (solo)	tookang://jobs/{job_id}
New job match (crew)	tookang://agency/jobs/{job_id}
Bid accepted	tookang://jobs/{job_id}/active
Crew invite received	tookang://crew/invite/{invite_token}
Dispute opened	tookang://disputes/{dispute_id}

13. Tech Stack

Layer	Technology	Notes
Web Frontend	Next.js 14 + Tailwind CSS	App Router; SSR for job/profile detail pages
Mobile	React Native (Expo)	EAS Build for production
Backend / API	Node.js + Express + Supabase Edge Fns	Edge Functions for AI and webhooks
Database & Auth	PostgreSQL (Supabase) + Supabase Auth	RLS policies enforced; separate views for solo vs agency data
Realtime	Supabase Realtime (WebSockets)	Chat, job status updates
AI	Claude API — claude-sonnet-4-20250514	Job assistant, matching, translation, photo analysis
Payments	Midtrans (Indonesia)	Mocked in MVP
Communications	Twilio / Wati WhatsApp Business API	Bridge + notifications
Storage	Supabase Storage	Photos, KTP, NIB/SIUP docs, PDFs
PDF Generation	Puppeteer (server-side)	Job Completion Certificates
Push Notifications	Expo Push / Browser Push API	Per Section 10
Voice Input	Web Speech API / Expo Speech	Bahasa Indonesia + English

14. Database Schema (Overview)

Updated in v1.5 to support the Solo Tukang / Mandor/Crew split with fully isolated rating columns.

users

id, email, phone, name, avatar_url, location, language_preference,
role (enum: user/tukang), account_type (enum: solo/mandor), created_at

tukang_profiles ● (Solo Tukang)

id, user_id (FK), bio, categories[] (max 3), service_area, rates,
solo_rating_avg, solo_jobs_count, solo_disputes_count,
verification_tier, ktp_verified (bool),
availability_status (enum: available/busy/on_leave),
bank_account_json, wallet_balance_idr, wallet_pending_idr

agencies ▲ (Mandor/Crew) ★ NEW

id, mandor_id (FK → users.id), agency_name, bio, logo_url,
trade_categories[] (all 8 possible), service_area,
agency_rating_avg, agency_jobs_count, agency_disputes_count,
nib_siup_doc_url, nib_verified (bool), ktp_verified (bool),
crew_availability_status (enum: available/busy/on_leave),
agency_wallet_balance_idr, agency_wallet_pending_idr,
agency_bank_account_json, created_at

agency_members ▲ ★ NEW

id, agency_id (FK → agencies.id), tukang_id (FK → tukang_profiles.id),
trade_specialty, joined_at, status (enum: active/inactive),
internal_score_avg (float — visible to Mandor only),
internal_score_count (integer)

jobs

id, user_id (FK), title, description, category, photos[], budget_min, budget_max,
urgency, status, location, requires_inspection (bool), inspection_fee_idr,
hired_type (enum: solo/crew), hired_tukang_id (FK, nullable), hired_agency_id (FK, nullable),

crew_member_ids[] (assigned members for this job), start_photo_url, finish_photo_url,
ai_price_estimate_json, photo_analysis_json, voice_transcript,
certificate_url, crew_welcome (bool), recommended_tier, created_at

bids

id, job_id (FK), bid_from_type (enum: solo/crew),
tukang_id (FK, nullable), agency_id (FK, nullable),
assigned_member_ids[] (for crew bids), amount, message,
duration_estimate, status, expires_at,
bid_type (enum: inspection_only/direct_quote), final_quote_amount

transactions

id, job_id (FK), user_id (FK),
payee_type (enum: solo/crew), tukang_id (FK, nullable), agency_id (FK, nullable),
total_amount, platform_fee_rate, platform_fee_amount, material_release_amount,
status (enum:
inspection_escrowed/full_escrow_funded/pending_confirmation/completed/disputed),
midtrans_order_id

reviews

id, job_id (FK), reviewer_id (FK),
review_target_type (enum: solo/crew),
tukang_id (FK, nullable), agency_id (FK, nullable),
on_time, price_accuracy, would_rehire, comment,
ai_authenticity_score, created_at

crew_member_internal_scores ▲ ★ NEW

id, job_id (FK), agency_id (FK), tukang_id (FK → crew member),
mandor_score (1–5 integer), mandor_note (text, optional), created_at
[Visible ONLY to the Mandor who owns the agency. Never exposed via public API.]

messages · notifications · disputes

Unchanged from v1.4. messages.thread_id links to the job record,
which carries hired_type so the UI can render solo vs crew chat context correctly.

15. MVP Scope vs Post-Launch

15.1 Hackathon MVP (v1.0)

Build the Solo Tukang flow end-to-end first — it is the simpler path and the majority use case.
Add Mandor/Crew as a demonstrable second tier — show the profile, bid card, and isolated rating UI.
Crew internal scoring can be mocked (static data) for the demo.

Priority	Feature	Tier
P0	User auth & role selection (solo / mandor path)	Both
P0	AI Job Posting + photo analysis + price estimate	Both
P0	Two-Stage Escrow State Machine	Both
P0	Bilingual Chat + WhatsApp Bridge (demo)	Both
P0	Solo Tukang profile, bidding, rating	Solo
P1	Mandor/Agency profile + crew member roster UI	Crew
P1	Crew bid card (distinct design from solo bid card)	Crew
P1	Separate rating display (solo ★ vs agency ★)	Both
P1	Voice input + guided review flow	Both
P2	Crew internal score (Mandor dashboard — can mock)	Crew
P2	Agency wallet (separate balance display)	Crew
P2	Job Completion Certificate PDF	Both

15.2 Post-Launch (v1.1 – v2.0)

- Full Smart Tukang Matching AI (dual-list: solo ranked + crew ranked)
- Dynamic badge system recalculated every 24 hours (both badge sets)
- Crew member invitation flow (WhatsApp link + in-app accept)
- Mandor internal team dashboard (crew member scores, assignment history)
- Map view for both solo and crew discovery
- Full Midtrans Payouts for both solo wallet and agency wallet
- In-app analytics for both Solo Tukang and Mandor (earnings, response rate)
- Agency leaderboards and verified agency directory page

16. Screen List

16.1 User Screens

- Onboarding (role selection, location, property type)
- Home Feed (recommended Solo Tukang + featured Mandor/Crew)
- Search (toggle: Solo / Crew / Both; filter by trade, area, rating, availability)
- Solo Tukang Profile View (solo badges, solo rating ★, portfolio)
- Agency Profile View (agency badge, agency rating ★, crew thumbnails, crew portfolio)
- AI Job Post (voice input, crew welcome toggle, photo upload, AI refine, dual price estimate)
- Bids Inbox (two tabs: Solo Bids / Crew Bids — or unified with type label)
- Escrow Pay (inspection fee) + Escrow Top-up (final quote)
- Active Job Chat (User ↔ Solo Tukang or User ↔ Mandor)
- Guided Review Screen (reviews the entity hired — solo or agency as a unit)
- Job Completion Certificate download

16.2 Solo Tukang Screens ●

- Onboarding (KTP, categories max 3, rates, bank account)
- Job Feed (solo-eligible jobs, AI fit score, crew-welcome indicator)
- Submit Solo Bid (inspection fee vs direct quote)
- Active Job (GPS check-in, start/finish photos, chat with User)
- Solo Wallet (balance, payout history, withdraw)
- Availability Calendar (personal status)
- My Crew Memberships (list of crews I belong to — read-only, leave option)

16.3 Mandor Screens ▲ ★ NEW

- Agency Onboarding (crew brand name, KTP, NIB/SIUP, trade categories, invite crew)
- Agency Job Feed (crew-eligible jobs sorted by AI fit score)
- Submit Crew Bid (select assigned members for this job, quote amount)
- Active Job — Mandor View (GPS check-in, start/finish photos, chat with User)
- Agency Wallet (agency balance separate from personal wallet, payout history)
- Team Dashboard (crew member roster, internal scores per job, assign to bid)
- Crew Invitation Manager (invite by phone, accept/pending/remove)

16.4 Admin Screens

- Dashboard (platform KPIs for both tiers: solo jobs vs crew jobs, escrow volume, disputes)
- Solo Tukang Verifications (KTP review, skill assessment, badge assignment)

- Agency Verifications (KTP + NIB/SIUP review, portfolio spot-check, Certified Agency approval)
- Disputes (AI summary view, resolution action — Mandor is sole named party for crew disputes)
- Transaction Monitor (separate ledgers: solo transactions vs agency transactions)

Appendix A — Changelog: v1.4 → v1.5

Section	Change	Type
3	User Roles expanded: Mandor and Crew Member added as distinct roles	NEW
4.1	Onboarding now branches into Solo vs Mandor paths (NIB/SIUP, crew invite)	UPDATED
4.2	Job posting: Crew Welcome toggle, AI tier recommendation	UPDATED
4.3	Bidding: separate Solo bid card and Crew bid card designs	UPDATED
4.4	Reviews: crew reviews go to agency rating only — solo profile never affected	UPDATED
5	NEW full section: Solo Tukang vs Mandor/Crew specification	NEW
5.1	Account type overview comparison table	NEW
5.2	Profile structure differences (solo vs agency)	NEW
5.3	Crew composition & member management rules	NEW
5.4	Bid card design differences	NEW
5.5	Rating system isolation — fully separate, never merged	NEW
5.6	Job-size eligibility routing via AI	NEW
5.7	Verification differences (KTP only vs KTP + NIB/SIUP)	NEW
5.8	Platform fee per tier (solo and agency independently)	NEW
5.9	Payout flow differences (personal wallet vs agency wallet)	NEW
7.1	AI Job Assistant now outputs recommended_tier	UPDATED
7.2	Smart Matching runs dual-pass: one for solo, one for crew	UPDATED
7.7	AI Price Estimator generates dual estimates (solo vs crew) for eligible jobs	UPDATED
9.2	Agency Wallet added as separate entity (min Rp 500K withdrawal)	NEW
11.2	Agency badge set (6 badges, fully independent from solo badges)	NEW
14	DB schema: agencies, agency_members, crew_member_internal_scores tables added	NEW
14	jobs.hired_type, bids.bid_from_type, reviews.review_target_type columns added	UPDATED
15	MVP priority table updated: P0/P1/P2 with tier labels	UPDATED
16.3	Mandor screen list added	NEW

— End of Document —